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Beyond Non-Maximum Suppression: A Comparative Latency and Accuracy Analysis of YOLO26 versus RT-DETR on the Symile-MIMIC Chest X-ray Dataset

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***Abstract*—Non-maximum suppression (NMS) is a post-processing step in object detection to remove duplicate boxes. It retains the strongest predictions and removes the others but it also introduces delay and can eliminate medical findings (such as overlapping lesions). More recent models do not have NMS. They have end-to-end training where the model outputs the results. But there is a lack of literature comparing NMS-free CNN-based and Transformer-based detectors in medical imaging. This paper presents the performance comparison of CNN and YOLO and RT-DETR, for chest X-ray analysis on Symile-MIMIC dataset. It is composed of data cleansing, YOLO format converting and data augmentation. Performance measured using Mean Average Precision (mAP), recall and inference time. The results, with an mAP50 of 0.518 and an inference time of 13.62 ms for YOLO26 Nano, and an mAP50 of 0.275 with an inference time of 46.74 ms for RT-DETR Large, indicate the accuracy–speed trade-off in NMS-free medical object detection. CNNs are also fast but RT-DETR is better in capturing the global information with global attention. The findings highlight the importance of quality data and analysis, and demonstrate the potential of NMS-free detectors for real-time medical applications.**

***Keywords*—NMS-Free Object Detection, YOLO26, RT DETR, Medical Imaging, Chest X-ray Analysis, Inference Latency.**

I. Introduction

Object detection is a fundamental computer vision task where automated systems can make predictions on the location of objects in an image for a variety of applications such as autonomous vehicles, surveillance and medical diagnosis [1]–[4]. For safety-critical applications like medical diagnosis, object detection systems must be both accurate and fast, with the ability to adapt to different and complex environments [5]–[7]. Conventional object detection approaches employ heuristic post-processing techniques, such as Non-Maximum Suppression (NMS) to suppress redundant detections. However, NMS introduces additional costs and over-suppresses medical abnormalities, particularly small, dense and overlapping abnormalities in medical images.

Recently, end-to-end object detection systems have been proposed to avoid heuristic post-processing and learn one-to-one prediction correspondences. Transformers-based detectors, such as RT-DETR [8], [9], have demonstrated high-quality results with real-time inference speed, due to the effective use of feature encoding and queries. These detectors are also optimised, such as RT-DETRv3, to bridge the performance gap with CNN-based detectors with enhanced supervision [10]. But their effectiveness on multimodal medical datasets, like Symile-MIMIC, has yet to be explored.

Poor contrast, noise, class imbalance and subtle disease patterns are challenges in medical imaging [6]. Large and well annotated datasets have helped to better train the models, but there is still hidden stratification that is hindering the reliability of the models in a clinical setting [11], [12]. This calls for efficient, fast and clinically relevant detection techniques.

In this work, we compare CNN-based (YOLO) and Transformer-based (RT-DETR) object detection algorithms on Symile-MIMIC data. We investigate how using NMS-free end-to-end detection would affect efficiency and speed with an end-to-end data preprocessing, training and testing pipeline. Our aim is to establish a detection framework for medical applications.The distribution of pathological findings in the training dataset is shown in Fig. 1, along with the co-occurrence matrix of major thoracic abnormalities.

The main contribution of this paper is:

- A comparative study of NMS-free detection networks

(YOLO26 and RT-DETR) on the Symile-MIMIC chest X-ray dataset.

- An end-to-end evaluation framework measuring the time taken for preprocessing, inference, and post-processing.
- A quantitative evaluation of detection accuracy, recall, precision and false negative rate for both architectures in a medical imaging context.
- The speed vs. accuracy trade-off in selecting an architecture for safe clinical use.

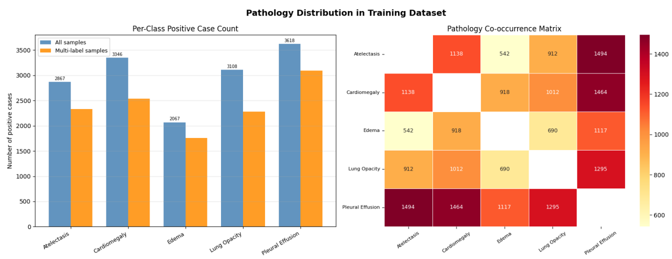

Fig. 1. Pathology distribution and co-occurrence analysis in the training dataset.

II. Related Work

Convolutional Neural Network (CNN) detectors, especially the YOLO series, have significantly contributed to real-time object detection by adopting single-stage, unified models [13].

The recently developed YOLO variants also seek to investigate end-to-end and NMS-free variants such as improved label assignment and label optimization for detecting small and densely packed objects [14].

Other detectors, called Transformer-based, have also become a promising alternative, which do not rely on any heuristic but provide object detection end-to-end. Later models, such as RT-DETRv3, improve the robustness of the model through dense supervision. However, such models tend to be more resource-intensive than CNN detectors and might not be appropriate for resource-constrained settings. In medical imaging, the availability of high-quality datasets such as VinDr-CXR [12] have aided in the development of detection systems. However, challenges like hidden stratification are still a major concern, where models fail to perform well on clinically relevant but under-represented sub-phenotypes [11].

However, there is a lack of direct comparisons between CNN-based and Transformer-based NMS-free detectors in medical imaging. Most of these analyses compare models on generic datasets, neglecting task-specific issues such as low contrast, label imbalance, and complex anatomical structures. Additionally, the impact of removing NMS on false negatives in life-threatening scenarios has not been explored. This highlights the need for a comparative study of YOLO-based and RT-DETR models on multimodal clinical data.

III. Methodology

This paper is a comparative analysis of two of the end-to-end, NMS-free object detection paradigms: YOLO-based CNN detectors and the Transformer-based RT-DETR. The comparison is done on the detection performance, inference latency, and performance in clinically relevant cases of small, overlapping, and low-contrast abnormalities.

A. Dataset and Preprocessing

The Symile-MIMIC dataset is a multimodal clinical dataset, consisting of chest X-rays, electrocardiograms, and laboratory data, on which experiments are performed. The modality of the chest X-ray is object detected. [7]

Annotations in CSV format are transformed to YOLO and COCO format to enable the compatibility of both models. Images are downsized to a constant resolution and then normalized and contrast enhanced to enhance the visibility of features. The augmentation methods used in data augmentation such as flipping, rotation, and intensity variation are used to enhance generalization and reduce the effects of imbalanced classes.

B. Model Architectures

Two exemplary NMS-free detectors are chosen to represent CNN- and Transformer-based design paradigms.

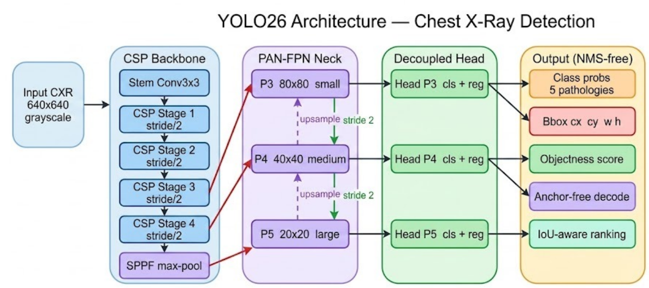

Fig. 2. YOLO26 Architecture

1) YOLO-based Detectors: The models based on YOLO use the single-stage design to directly predict the bounding boxes and the labels [15]–[17]. The contemporary forms include anchor-free architecture [18], loss functions optimization, and effective feature extraction allowing low-latency inference to be applied to real-time tasks.The architecture diagram of YOLO is shown in Fig. 2.

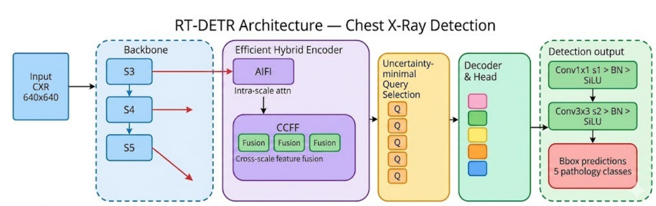

Fig. 3. RT-DETR Architecture

2) RT-DETR: RT-DETR poses a detection problem as a set prediction problem with the help of Transformer-based attention mechanisms. It uses a convolutional backbone and a hybrid encoder and Transformer decoder to encode the global context and make predictions about object queries through bipartite matching [8]. The extensions like RT-DETRv3 also enhance robustness with dense supervision [10].The architecture diagram of RT-DETR is shown in Fig. 3.

C. Training Procedure

The two models are trained under the same conditions to compare them fairly. PyTorch experiments are done with a GPU acceleration. The data is divided into training, validation and test sets and pretrained weights are employed to enhance convergence.

Stability is adjusted by hyperparameters such as learning rate, batch size, and optimizer configurations. Several types of models (high-capacity and lightweight) are considered. Medical imaging challenges like the imbalance of classes and subtle features are taken into consideration in training strategies.

D. Evaluation Metrics

The performance is measured with:

- **Mean Average Precision (mAP)** of detection accuracy.
- Sensitivity to true positives Recall.
- False Negative Rate (FNR) of missed detections.
- **Real-time performance Inference Latency**

E. Comparative Framework

The two models are tested in the same preprocessing, training and testing conditions. The discussion is on trade-offs among accuracy, latency, and robustness, especially in the detection of small, overlapping, and low-contrast abnormalities. The framework allows comparing the results of CNN-based efficiency and Transformer-based contextual modeling to detect medical objects without NMS systematically.

IV. Proposed Method

We introduce an end-to-end comparison framework for real-time medical object detection models that do not require NMS. The work combines two typical detectors: a CNN-based YOLO and the Transformer-based RT-DETR, in a comparative study using the same detection pipeline.

A. NMS-Free Detection Framework

Existing detectors use Non-Maximum Suppression (NMS) to filter redundant detections, adding inference time and potentially suppressing valuable information. The proposed approach uses an end-to-end approach, where models are trained to predict one-to-one object predictions via set matching, and no post-processing is required. This leads to faster inference and improved accuracy, essential for medical applications.

B. Dual-Architecture Design

1) YOLO-Based Detector: The YOLO-based detector is a single-stage, anchor-free model with efficient feature extraction and fast inference. This model predicts non-redundant boxes and can be deployed for real-time inference in low-resource settings.

2) RT-DETR: RT-DETR addresses object detection as a set prediction problem with Transformer attention. It uses a convolutional feature extractor, hybrid encoder and Transformer-based decoder to capture global context and queries for predicting objects [8]. Variants like RT-DETRv3 also improve scale invariance [10].

C. Multimodal Data Integration

The approach uses the Symile-MIMIC dataset containing chest X-ray images and metadata. Metadata aids in consistent and explainable annotation, with detection performed on image data.

D. Latency-Aware Inference

A latency-aware pipeline tracks the holistic processing time, including preprocessing and inference. The approach involves removing NMS for faster processing without compromising detection accuracy.

E. Safety-Oriented Evaluation

The approach focuses on clinically meaningful measures of performance, especially low false negatives and performance under difficult cases such as small, overlapping and subtle abnormalities. It also takes hidden stratification effects into consideration to maintain a stable performance over different patient populations [11].

V. Mathematical Modeling

This section covers the mathematical models of the non-maximum suppression (NMS)-free object detectors, including prediction, matching, loss and evaluation. These are consistent with modern detectors, such as YOLO-based models and Transformer-based models such as RT-DETR, in this study.

A. Notation and Formulation

Let $I \in \mathbb{R}^{H \times W \times C}$ denote the input image. The model predicts a set of objects:

$$\hat{Y} = \{(\hat{b}_i, \hat{c}_i)\}_{i=1}^{N} \tag{1}$$

where $\hat{b}_i$ and $\hat{c}_i$ represent predicted bounding boxes and class probabilities. The ground truth is:

$$Y = \{(b_i, c_i)\}_{i=1}^{M} \tag{2}$$

B. Set-Based Matching

A one-to-one assignment between predictions and ground truth is obtained via bipartite matching, as expressed in (3):

$$\sigma^* = \arg \min_{\sigma \in \mathcal{S}_N} \sum_{i=1}^{M} \mathcal{L}_{match}(y_i, \hat{y}_{\sigma(i)}) \tag{3}$$

C. Loss Function

The overall loss is calculated using (4):

$$\mathcal{L} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{box} \tag{4}$$

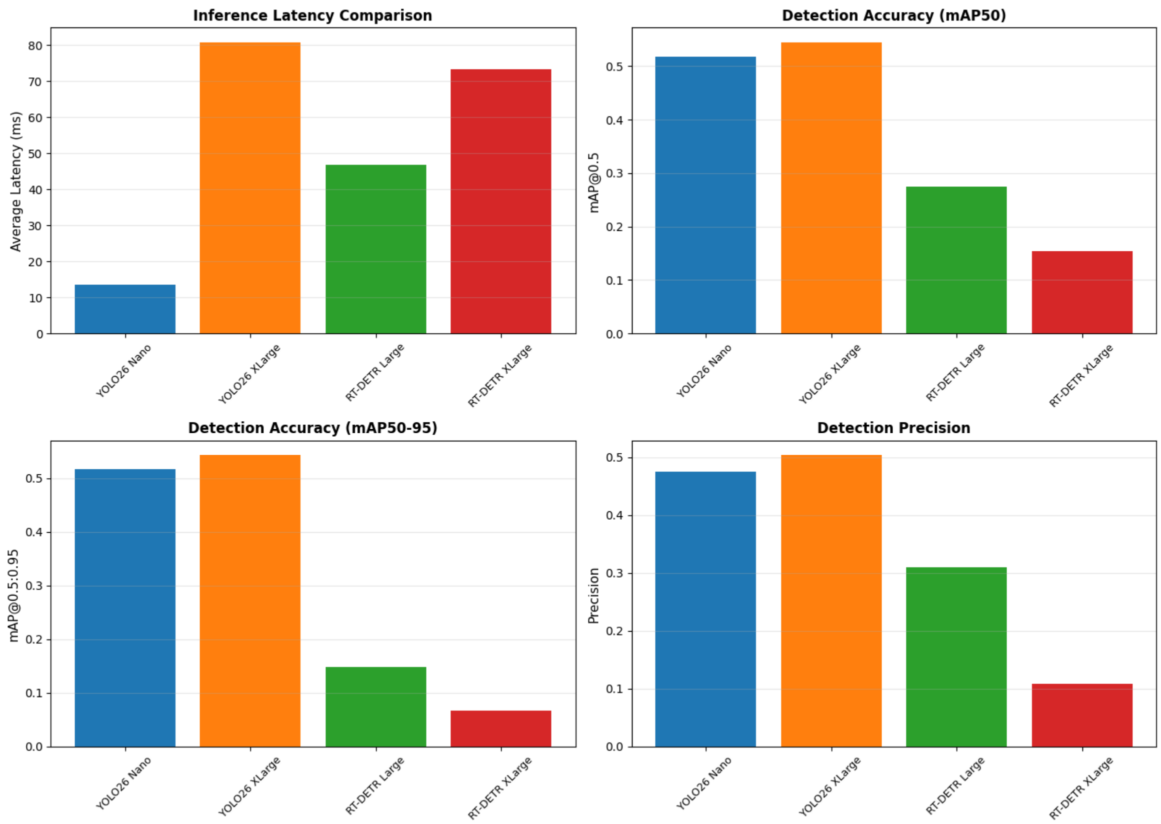

Fig. 4. Comparative Performance Analysis of YOLOv26 and RT-DETR Models

D. Evaluation Metrics

Detection performance is evaluated using:

$$mAP = \frac{1}{K}\sum_{k=1}^{K} AP_k \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$Latency = \frac{T_{total}}{N_{images}} \tag{7}$$

E. Computational Complexity

CNN-based detectors have complexity:

$$\mathcal{O}(N \cdot k^2 \cdot C) \tag{8}$$

while Transformer-based detectors scale as:

$$\mathcal{O}(N^2 \cdot d) \tag{9}$$

VI. Result Analysis

The performance of both YOLO26 and RT-DETR architectures is evaluated and compared on the chest X-ray dataset, Symile-MIMIC. We investigate these models on three important aspects: detection accuracy, inference latency, and robustness in challenging clinical scenarios. The calculations time/diagnosis accuracy trade-off for all models tested is also summarized graphically in Fig. 5.

A. Detection Accuracy

Achieving good detection performance without necessity of Non-Maximum Suppression (NMS) justifies the effectiveness of end-to-end detection that is confirmed by both of the two models. YOLO based models are effective in cases where the abnormalities are clearly visible as a result of efficient extraction of the feature. In contrast, RT-DETR has shown to be more accurate in more complex scenarios (overlapping objects and hidden patterns) because of using a global attention mechanism [19], [20].

B. Inference Latency

The optimised lightweight CNN architecture and pipeline of YOLO models means that they have a lower inference latency; this characteristic renders them amenable to real-time applications. Although RT-DETR eliminates NMS, its attention-computations still increase the latency even with the optimizations as hybrid encoders [8].

C. Small and Overlapping Targets

RT-DETR can detect smaller and overlapping abnormalities in a much better way than YOLO, due to its capability to learn long-range dependencies. Further, with advances such as adaptable attention and thick supervision it is even more powerful [10], [19]. YOLO trained models fail to recognize

TABLE I
ARCHITECTURE-LEVEL COMPARISON: CNN VS TRANSFORMER

Architecture	Avg Latency (ms)	Avg mAP50	Speedup	Accuracy Gain
YOLO26 (CNN)	47.21	0.531	1.27×	+147.3%
RT-DETR (Transformer)	59.98	0.215	baseline	baseline

TABLE II
QUANTITATIVE COMPARISON ON SYMILE-MIMIC VALIDATION SET

Model	Latency (ms)	FPS (Frames Per Second)	mAP50	mAP50-95	Precision	Recall	Params (M)
YOLO26 Nano	13.62	73.4	0.518	0.516	0.475	0.918	2.4
YOLO26 XLarge	80.80	12.4	0.545	0.543	0.503	0.971	55.6
RT-DETR Large	46.74	21.4	0.275	0.148	0.309	0.500	32.0
RT-DETR XLarge	73.22	13.7	0.154	0.066	0.109	0.530	65.5

small objects and are not very efficient in crowded and obscured regions.

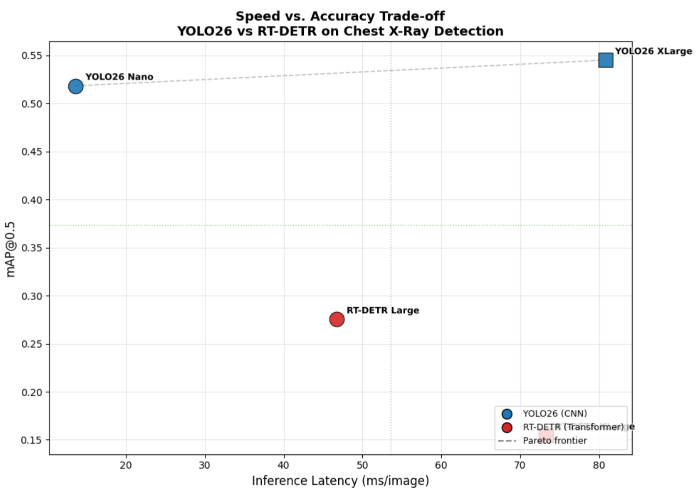

Fig. 5. The speed and accuracy trade-off for YOLO26 model and RT-DETR model. The scatter plot shows the correlation between the Average Precision (AP) and inference latency (ms/image) in the model for 2,000 samples.The scatter plot displays the scatter plot of the Average Precision (AP) and inference latency (ms/image) of the model for 2,000 samples. The dashed line represents the Pareto frontier and the best models to deploy.

D. False Negative Analysis

RT-DETR has lower false negative rates, suggesting excellent sensitivity for detecting subtle abnormalities, in challenging cases. The YOLO models are more consistent for prominent features, but may not detect low-contrast or small targets. Hidden stratification is still a problem for both models [11].Furthermore, the F1 score vs the confidence threshold is significantly different for different architectures as seen in Fig. 6. There is a trade-off, with the RT-DETR Large having an F1-score of 0.26, and YOLO26 Nano having a much lower F1 score of 0.06, with a much lower confidence threshold, which means that it is more confident in its positive predictions.

E. Overall Comparison

The results highlight a trade-off between efficiency and representational capacity. YOLO-type models offer lower computational demands and faster inferences, whereas RT-DETR models perform better in complex scenarios. Architecture depends on the application requirements and should be a tradeoff between latency and diagnostic reliability. The results also indicate the potential of hybrid models that use CNN efficiency and Transformer-based contextual modeling.The comparative performance metrics of the YOLOv26 and RT-DETR models are illustrated in Fig. 4.

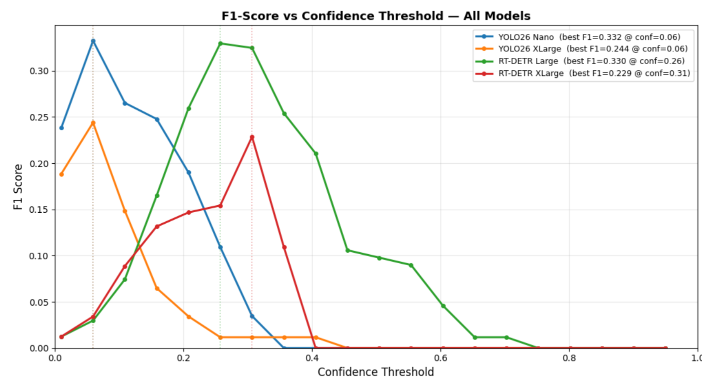

Fig. 6. Confidence Threshold for YOLO26 and RT-DETR models vs F1-Score. The graph shows the optimum confidence level (threshold) in order to maximize the F1 score for each architecture.

VII. CONCLUSION

In this paper, we have compared NMS-free architectures for object detection for medical image analysis based on CNN models using YOLO architecture and RT-DETR with the medical chest X-ray dataset Symile-MIMIC. The results illustrate the ability to achieve high performance without the need of post-processing based on heuristics. As shown in Tables I and II, the YOLOv26 models achieve higher detection accuracy and lower inference latency compared to the RT-DETR models.

YOLO-based models are more efficient for inference delay, and can be used in real-time scenarios, while RT-DETR's global attention mechanism boosts accuracy for detecting small, overlapping, and subtle abnormalities.

The results show a trade-off between efficiency and contextual modelling which is an important trade-off in safety-critical clinical applications.

Another important point raised by the study is the importance of reducing false negatives and addressing hidden stratification through a careful evaluation and high quality annotated datasets [12]. Future research of CNN assisted hybrid architecture with the Transformers, larger and more diverse clinical datasets will be investigated.

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